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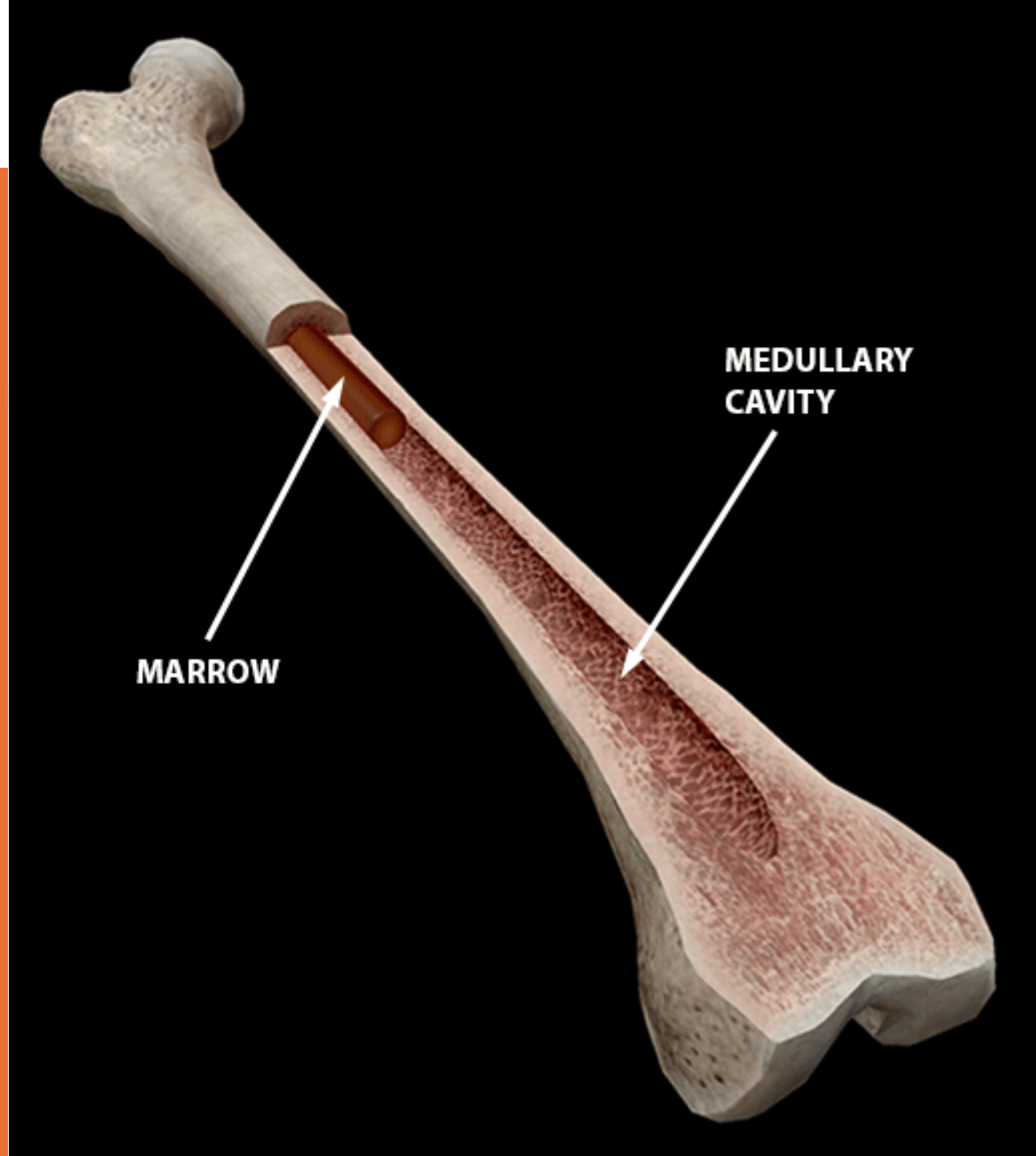
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Bone

- **Defined as:**

calcified connective tissue consisting of cells (osteocytes) embedded in a matrix of ground substance and collagen fibres, have a superficial thin layer of compact bone around a central mass of spongy bone, and contain internal soft tissue, the marrow, where blood cells are formed



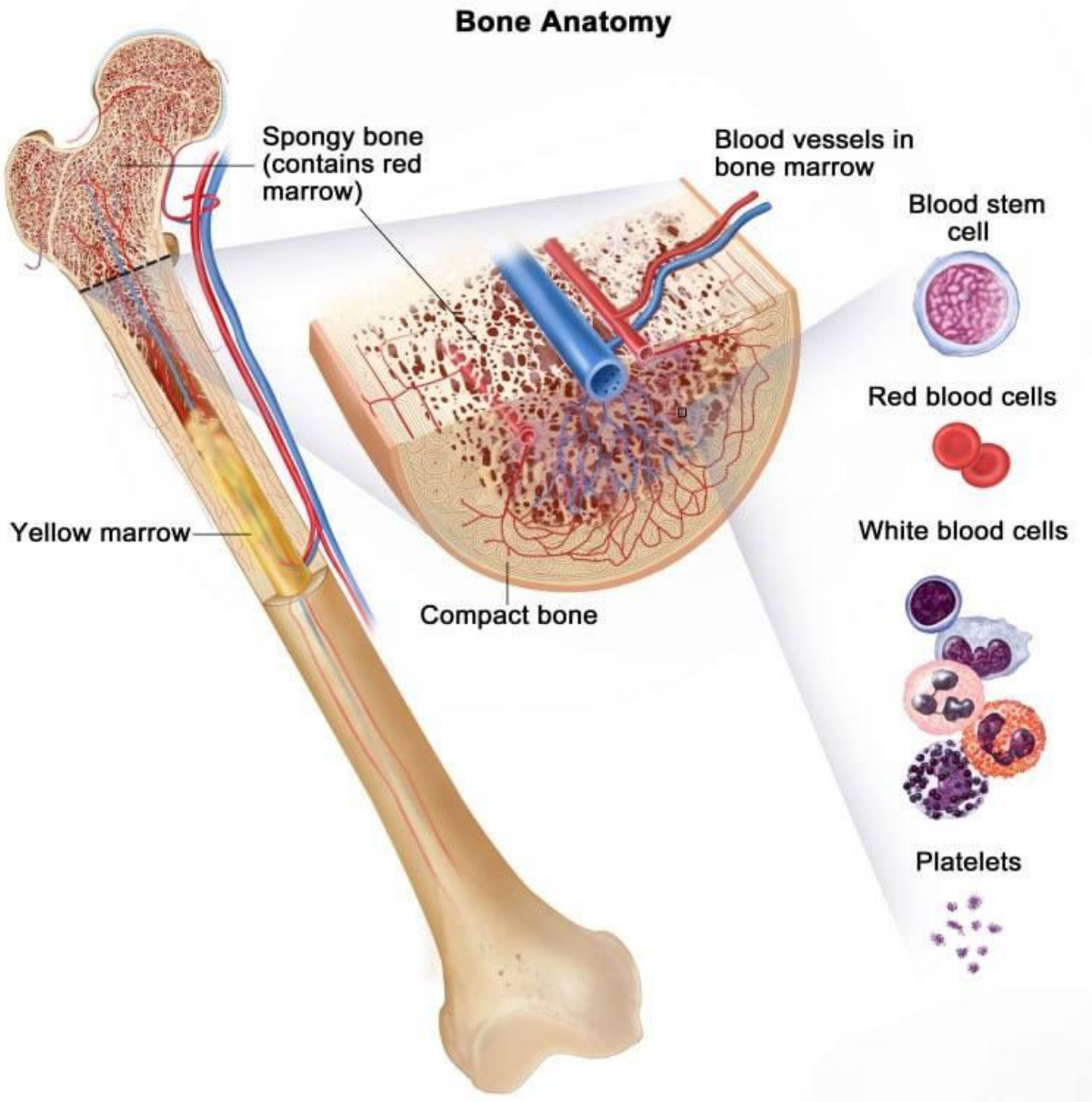
Bones are classified according to:

1. Structure into compact and spongy.
2. Position into axial skeleton (bones of head, vertebral column and sternum) and appendicular skeleton (bones of upper and lower extremities)
3. shape, into long, short, flat, irregular, and sesamoid bones
4. developmental history, into endochondral and membranous bones

Types of bones according to structure:

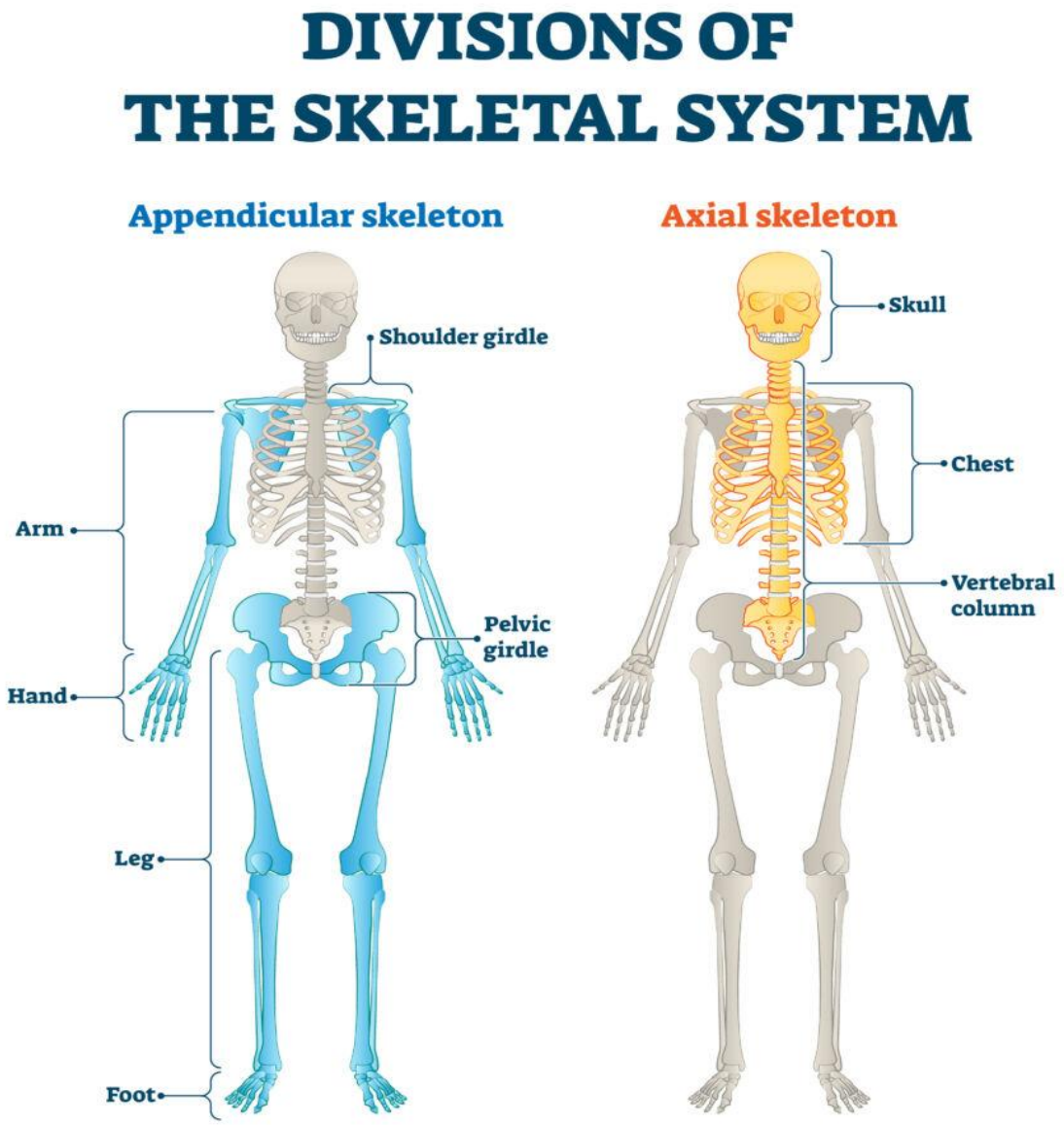
There are two types of bone tissue: compact and spongy.

The names imply that the two types differ in density, or how tightly the tissue is packed together



Types of bones according to position:

axial skeleton (bones of head, vertebral column, ribs and sternum)
appendicular skeleton (bones of upper and lower extremities).



CLASSIFICATION OF BONES



(According to Shape)

1. Long Bones

- *Shape: Longer than wide*
- *Have: Shaft & two ends*
- *Function: Movement & support*

Examples:
"Long = Limb bones"



Femur *Humerus* *Tibia*

2. Short Bones

- *Shape: Cube-shaped*
- *Function: Stability & little movement*

Examples:
"Short = Small & Strong"



Carpals *Tarsals*

3. Flat Bones

- *Shape: Thin & flat*
- *Function: Protection of organs*

Examples:
"Flat = Protect"



Sternum *Scapula* *Parietal Bone*

4. Irregular Bones

- *Shape: Complex shape*
- *Function: Protection & support*

Examples:
"Irregular = Weird Shape"



Vertebrae *Sphenoid*

5. Sesamoid Bones

- *Found in: Tendons*
- *Function: Reduce friction*

Example:
"Sesamoid = In Tendon"



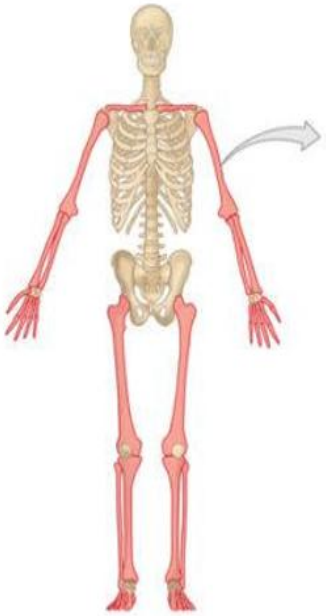
= *Patella*
(Kneecap)

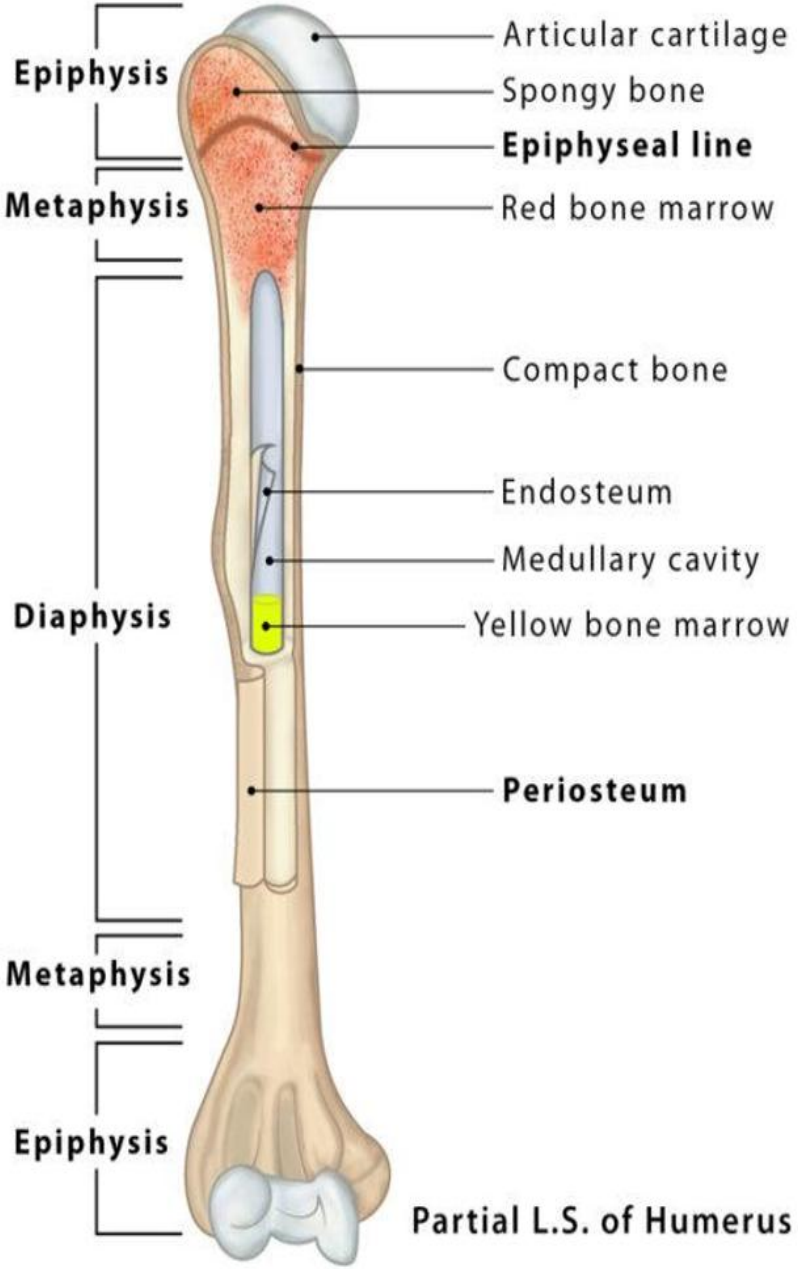
Extra: Rare Types



Sutural Bones *Pneumatic Bones*

Long Bone




Epiphysis
Metaphysis
Diaphysis
Metaphysis
Epiphysis
Articular cartilage
Spongy bone
Epiphyseal line
Red bone marrow
Compact bone
Endosteum
Medullary cavity
Yellow bone marrow
Periosteum
Partial L.S. of Humerus

Gross anatomy of long bone:

A long bone has two parts: the diaphysis and the epiphysis.

- The diaphysis is the tubular shaft that runs between the proximal and distal ends of the bone. The hollow region in the diaphysis is called the medullary cavity, which is filled with yellow marrow. The walls of the diaphysis are composed of dense and hard compact bone

description: each long bone has a shaft (diaphysis) and two ends (epiphyses). The metaphysis is a part of the diaphysis adjacent to the epiphyses.

1. Diaphysis: Forms the shaft (central region) and is composed of a thick tube of compact bone that encloses the marrow cavity.

2. Metaphysis: Is a part of the diaphysis, the growth zone between the diaphysis and epiphysis during bone development.

3. Epiphyses: Are expanded articular ends, separated from the shaft by the epiphyseal plate of hyaline cartilage during bone growth and composed of a spongy bone surrounded by a thin layer of compact bone

Structure of growing long bone:

Growing long bone presents:

- Shaft (diaphysis)
- Ends (epiphyses)
- Epiphyseal cartilage
- Epiphyseal end of shaft (metaphysis)
- Outer periosteum
- Inner endosteum
- Marrow cavity in diaphysis

وَلَقَدْ خَلَقْنَا الْإِنسَانَ مِن سُلَالَةٍ مِّن طِينٍ (12) ثُمَّ
جَعَلْنَاهُ نُطْفَةً فِي قَرَارٍ مَّكِينٍ (13) ثُمَّ خَلَقْنَا النُّطْفَةَ
عَلَقَةً فَخَلَقْنَا الْعَلَقَةَ مُضْغَةً فَخَلَقْنَا الْمُضْغَةَ عِظَامًا
فَكَسَوْنَا الْعِظَامَ لَحْمًا ثُمَّ أَنشَأْنَاهُ خَلْقًا آخَرَ ۚ فَتَبَارَكَ
اللَّهُ أَحْسَنُ الْخَالِقِينَ (14) ثُمَّ إِنَّكُم بَعْدَ ذَٰلِكَ لَمَيِّتُونَ
(15)